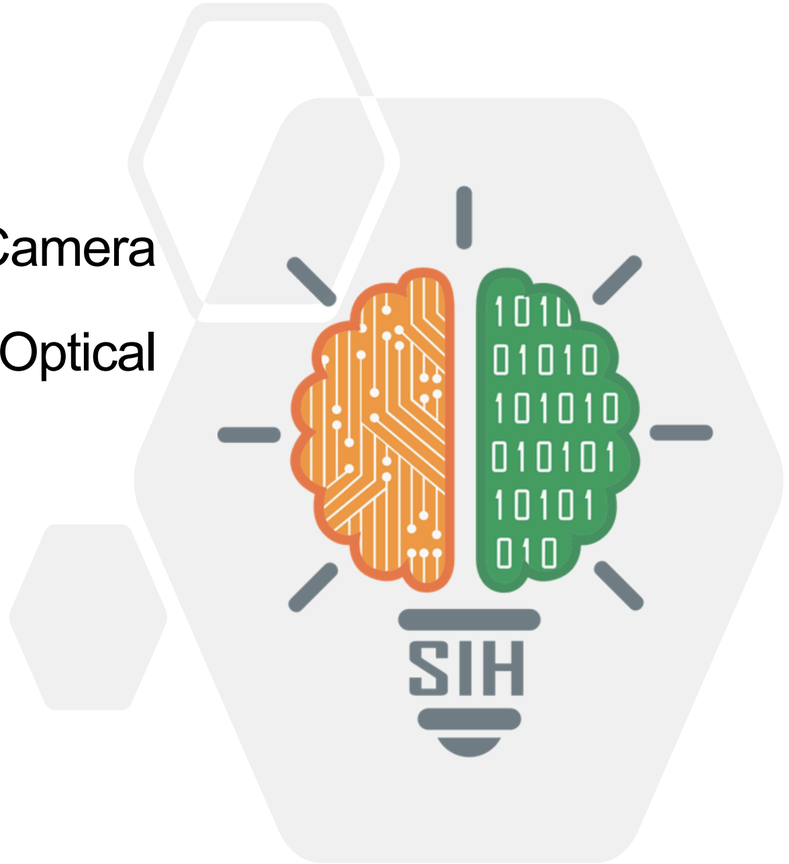
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SkyLock

- **Problem Statement ID -** 26169
- **Problem Statement Title -** Development of an AI-Based Virtual Camera Tracking System for Coarse Alignment of Mobile Free Space Optical Communication (FSOC) Terminals
- **Theme-** Smart Automation
- **PS Category-** Software
- **Team ID-** 186359
- **Team Name -** Nirvana



Nirvana

SkyLock

Run here: https://sky-lock.vercel.app/

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Proposed Solution

A software-simulated AI camera tracking system that automatically detects and continuously tracks a moving optical beacon between mobile FSOC terminals, enabling accurate coarse pointing without real hardware.

How It Works

1. **Simulate** — Virtual scene with moving satellite beacon
2. **Capture** — Virtual camera observes scene (pan/tilt control)
3. **Detect** — Computer vision locates beacon in each frame
4. **Predict** — Kalman filter smooths and predicts position
5. **Control** — PID computes pan/tilt correction
6. **Track** — Camera repositions and repeat in real time
7. **Add Realism** — Inject turbulence, vibration, noise
8. **Evaluate** — Live dashboard and auto-generate performance log



How It Addresses the Problem

- **Software-only testbed** — Removes hardware cost & complexity
- **Validate algorithms** — Reusable and configurable environment
- **Reliable acquisition** — Works under realistic disturbances

Innovation & Uniqueness

- **Hybrid approach** — CV + Kalman (+ optional lightweight DL)
- **Configurable disturbances** — Adjustable intensity (turbulence, vibration, noise)
- **Real-time analytics** — Metrics aligned with real PAT evaluation
- **Modular design** — Easily extendable to real hardware

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TECHNICAL APPROACH

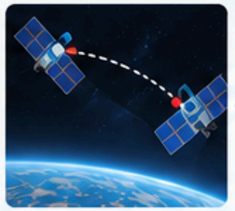
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Technologies to be Used

- **Python** — Core development
- **OpenCV (cv2)** — Detection, blob/contour analysis, frame processing
- **Kalman Filter** — Motion prediction & smoothing
- **PID Controller** — Camera pan/tilt correction loop
- **Pygame / OpenCV** — 2D simulation & virtual camera rendering
- **YOLOv8-nano** — DL-based detector (stretch goal)
- **PyQt / Tkinter / Flask** — Live dashboard / GUI
- **PyInstaller** — Standalone executable

Methodology and Process Flow

1 Virtual Scene Generation



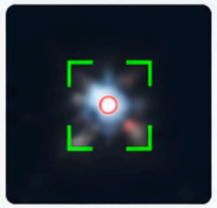
- Create simulated environment
- Moving beacon (configurable motion pattern)

2 Virtual Camera Capture



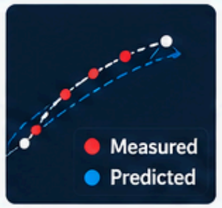
- Capture frame within FOV
- Pan/tilt control

3 Target Detection



- Detect beacon (OpenCV / YOLO)
- Blob / contour analysis

4 Kalman Filter



- Predict next position
- Smooth noisy measurements

5 PID Control



- Compute pan/tilt correction
- Keep target centered

6 Camera Reposition



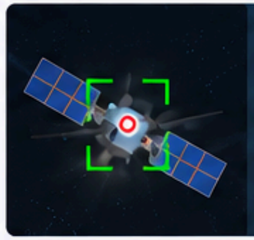
- Update pan/tilt angles
- Repeat every frame (real time)

Disturbance Module

Atmospheric turbulence · Platform vibration · Sensor noise · Blur & scintillation

Configurable disturbance intensity to test robustness

Output & Monitoring



Tracking	: LOCKED
FPS	60
Acquisition Time	1.2 s
Tracking Error	0.08°
Lock Retention	97.6%

Live dashboard (real-time stats) · Auto log for performance report

Outcome

Reliable coarse alignment in simulation

- Demonstrates robust tracking
- Handles real-world disturbances
- Reusable testbed for algorithm development

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FEASIBILITY AND VIABILITY

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TECHNICAL FEASIBILITY

Software simulation
Virtual environment eliminates need for expensive optical hardware.

Mature CV pipeline
Beacon detection and tracking achievable using OpenCV and existing AI models.

Proven algorithms
Kalman filtering and PID control are well-established and reliable.

Modular and extensible
Designed to integrate with real hardware in future.

Executable prototype
Can be developed and packaged as a standalone application.

KEY RISKS

Detection loss
Beacon may be outside FOV or obscured by noise/disturbance.

Environmental disturbances
Turbulence, vibration, blur and sensor noise can degrade tracking accuracy.

Processing overhead
Real-time detection + tracking + control must maintain high FPS.

Model limitations
CV/AI models may fail under extreme conditions (high noise, blur).

Simulation-to-reality gap
Differences when transitioning to real hardware.

MITIGATION

CV + Kalman (and optional DL)
Robust detection and prediction of target position.

PID control
Stable and smooth camera adjustments.

Re-acquisition logic
State machine for automatic recovery after lock loss.

Configurable simulation
Adjustable disturbance levels and scenarios for stress testing.

Optimized pipeline
Efficient processing for real-time performance (high FPS).

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IMPACT AND BENEFITS

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Impact on the Target Audience

Unjammable, Uninterceptable Links
Laser beams can't be intercepted or jammed like RF broadcasts.

Built for the Gb→Tb Era
Powers the pointing backbone behind gigabit-to-terabit FSOC speeds.

Faster, Cheaper R&D Cycle
Validates tracking algorithms in simulation before costly hardware.

A Reusable Tracking Core
Same detect-predict-control loop applies to debris, drones, and telescopes.

Key Benefits of the Solution

Spectrum-Free Scalability
Runs on license-free spectrum, sidestepping the RF crunch.

Economic
Eliminates costly cameras and pan-tilt hardware.

Strategic — Cuts Import Dependency
Supports India building lasercomm capability domestically.

Technical / R&D
Reusable testbed to benchmark algorithms under identical conditions.

Riding a Fast-Growing Market
FSO communications are on a steep multi-year growth curve.

Educational
Hands-on way to learn real-world PAT concepts.

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RESEARCH AND REFERENCES

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- **A Review on Pointing, Acquisition, and Tracking Approaches in UAV-based FSO Communication Systems," Optical and Quantum Electronics, Springer -** https://link.springer.com/article/10.1007/s11082-022-03968-2
- **Laser Based Free Space Optical Communication System for Future Space Application" (IEEE) — studies Atmospheric Modelling, ATP, and link budget within the framework of ISRO's satellite capabilities -** https://ieeexplore.ieee.org/document/9484908/
- **OpenCV Kalman Filter documentation — motion prediction and smoothing used in our tracking approach -** https://docs.opencv.org/4.x/dd/d6a/classcv_1_1KalmanFilter.html